Natural Capital and the Measurement of Productivity*

Alexander C. Abajian†, Marc N. Conte‡, and Eli P. Fenichel§

July 13, 2026

Abstract

Omitting inputs to production, such as natural capital, affects estimates of productivity growth. We show that benchmark measurements of productivity growth are strongly correlated with environmental change at the national level, consistent with biased measurement due to the omission of natural capital. We derive theoretical conditions under which the measurement and inclusion of previously-omitted environmental inputs reduce bias and error when estimating productivity growth. Calibrated simulations using data from the World Bank and Penn World Tables suggest including non-renewable natural capital in the measurement of TFP growth increases accuracy for 94 of the 100 countries in our sample.

Keywords: Total Factor Productivity; Natural Capital; Economic Measurement

JEL Codes: E01, O44, O47, Q56

*We thank Leonard Nakamura, Scott Wentland, Elizabeth Kiser, David Keiser, Javier Birchenall, Henry Kim, Luis Diego Herrera Garcia, Dale Manning, and seminar participants at the Yale School of the Environment, AERE sessions at WEIA, and the 27th annual BIOECON conference for their feedback. Abajian and Fenichel gratefully acknowledge financial support from the Knobloch Family Foundation. All errors are our own.

†Abajian: Yale School of the Environment, alexander.abajian@yale.edu

‡Conte: Fordham University, marc.conte@fordham.edu

§Fenichel: Yale School of the Environment, eli.fenichel@yale.edu.

Innovation and technological change drove an unprecedented increase in economic activity over the past two centuries. At the same time, industrialization muddled societal awareness of fundamental connections between humans and nature. Society's interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital. Then-President Theodore Roosevelt noted this phenomenon at the outset of the 20th century:

> "With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature." – Roosevelt (1908)

Formal systems of economic measurement followed this trend; today's structures provide little guidance for separating nature's contribution to production from the contributions of other factors. Starting with Simon Kuznets' report to Congress in 1934, the broader development of statistical, economic, and environmental policy has focused on measuring output and productivity growth by tracking produced capital and labor over time (Kuznets 1934). While governments historically emphasized tracking natural resources for posterity and avoiding overuse, features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today. For example, sailors, vessels, and sonar are essential inputs to industrial fishing and are measured by national accountants on a regular basis; equally important is the fish stock and its ability to replenish itself over time (Hannesson 2007). However, the fish stock as an input to production is often excluded from economic measurement. This continues despite managed fish stocks (within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts (United Nations 2009, 2025).

Here, we examine how the omission of the environment as a factor in production affects a leading measure of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP growth is defined as the growth in output unexplained by growth in inputs, and it has been a benchmark metric for economy-wide technological progress and innovation for nearly 70 years (Solow 1957). A country's level of TFP along with its capital stock is a sufficient statistic for its residents' welfare (Basu et al. 2022). Forecasts of TFP growth play a crucial role in shaping governments' budget projections and policy agendas over long time horizons (CBO 2025). If natural capital affects production, then the omission of environmental assets from national accounting frameworks induces error when estimating aggregate productivity growth. Failing to account for how changes in the environment affect output may lead governments to overestimate the ability of society to grow without the use of nature or to misattribute the effects to economic policy (Nordhaus and Tobin 1973; Brandt, Schreyer, and Zipperer 2017).

In this paper, we first demonstrate how omitting inputs, such as natural capital, from the production function affects residual-based estimates of TFP and discuss why natural capital is often omitted in practice. We provide predictions for the correlation of environmental change with productivity growth when natural capital is incorrectly omitted from the set of measured inputs used to construct TFP. Then, we test whether a benchmark series for TFP growth is orthogonal to changes in natural capital services in a balanced panel of data covering 117 countries between 1996 and 2019. We find measured growth is strongly correlated with changes in the modest set of natural capital stocks we consider. The correlations between productivity growth and natural capital persist when using alternative productivity measurements from the economics literature or those calculated by national statistical agencies.

Next, we characterize when inclusion of natural capital aids measurement of TFP in theory and consider whether it does so in practice. We derive conditions under which additional measurement unambiguously reduces bias in residual-based estimates of TFP growth. These conditions prove to be restrictive and unlikely to hold in most settings. As the theoretical conditions are not formally testable, we use simulation-based evidence to gauge whether including natural capital is likely to improve measurement empirically. The results suggest that including natural capital produces gains in measurement accuracy. Across simulations calibrated to data from 100 countries, the median level of increase in accuracy, measured as the reduction in root mean squared error for estimated average TFP growth, is 7.3 basis points, or 12.5 percent of the average TFP growth across all countries during the sample period. To give a sense of the scale of this error, were this the estimate for the United States, and were it componded over a standard 10-year budget horizon, then the resulting forecast error for cumulative revenues would be a little over \$300 billion in 2025.[1] This figure is two to three times larger than the annual year-on-year forecast error in deficit projections by the U.S. Congressional Budget Office (CBO), or roughly a quarter of the aggregate cost of compliance with the Clean Air Act between 1970 and 1990 (CBO 2026; U.S. Environmental Protection Agency 1997).

Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value as part of a broader measure of economic welfare as well as the productive value of land (Nordhaus and Tobin 1973). Subsequent research incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to produced capital (e.g., Mankiw, Romer, and Weil 1992; Caselli and Feyrer 2007; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis 2019). These studies apportion capital income between returns on investments in produced capital and resource rents. Broadly, the literature has highlighted the importance of correcting for resource rents when comparing the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains substantial (Hsieh and Klenow 2010). We add to this literature by characterizing how introducing previously omitted factors of production has ambiguous

1. This cumulative deviation would grow to \$1.2 trillion if we were to instead apply our U.S.-specific point estimate for the improvement in RMSE induced by including non-renewables on the margin.

effects on the bias of measured output elasticities and factor shares.

A more nascent strain of literature focuses on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to those by Brandt, Schreyer, and Zipperer (2017) and Manning and Barbier (2025), who study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuels. Related work by Freeman, Inklaar, and Diewert (2021) highlights the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular measures of TFP. We add to this literature by explicitly characterizing the conditions under which measurement of non-renewable resources reduces errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. Our approach can also be generalized to consider the effects of including other potentially omitted input stocks, such as renewable natural capital.

1 Residual-Based Productivity Measurement

Neoclassical economics attributes output to two sources: factors of production (factor inputs) and technology. Final output at each instant, $Y(t)$, is determined by two objects: the level of factor inputs at time t, denoted by the set $\boldsymbol{X} \triangleq \{X_1(t), \ldots, X_n(t)\}$, and a time-specific summary statistic for the level of technology available to society, total factor productivity, $TFP(t)$:

$$Y(t) = TFP(t) \times F(\boldsymbol{X}). \tag{1}$$

TFP is a theoretical modeling device. It is an index number capturing how technology determines the production possibility frontier at a given point in time.[2] It is a parsimonious treatment for how forces such as innovation and creative destruction shift what society can produce given available resources (Romer 1990; Aghion and Howitt 1992). Under standard assumptions for the production function (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output is given by

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \boldsymbol{X}} \gamma_x g_x(t), \tag{2}$$

where $g_x(t)$ terms denote growth rates and γ_x are the output elasticities for each input x at time t.[3,4] The growth accounting equation in (2) states that all changes in output are attributable to factors of production or growth in TFP (Barro 1999). Rearranging the growth accounting equation

2. We assume throughout this paper that technical change is Hicks neutral. A more general formulation of technical change is the partial derivative of the function determining aggregate output with respect to time.

3. Here we refer to the γ_x terms as output elasticities. Section 1 of the online appendix shows that all analysis holds when the γ_x terms are reinterpreted as the income shares for factors of production.

4. We assume throughout that output elasticities γ_x are time-invariant.

gives

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \boldsymbol{X}} \gamma_x g_x(t). \tag{3}$$

TFP growth is the output growth unexplained by growth in inputs (Comin 2010). The accounting identity in (2) requires that these two concepts of productivity growth—the growth rate of an index number for technology and the residual portion of growth unexplained by factor inputs—coincide.

In practice, TFP growth is measured using the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\boldsymbol{I}$. Factor inputs (e.g., produced capital and labor) and final output are measured by national statistical agencies at regular frequencies, allowing for statisticians to take versions of (3) to the data. Residual changes in observed output—those unexplained by growth in observed inputs—are called Solow residuals (Solow 1957) and are imputed to productivity growth by solving

$$g_A(t) \triangleq \hat{g}_Y(t) - \sum_{i \in \boldsymbol{I}} \hat{\gamma}_i \hat{g}_i(t), \tag{4}$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are estimates of growth and output elasticities, respectively. A Solow residual is an estimate of TFP growth. Versions of equation (4) are used by national statistical agencies to construct productivity growth in national accounts (Meyer and Harper 2005; European Commission and Eurostat 2025), and in academic applications comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer 2015).

If national accountants were able to calculate the dynamics of all aspects of the economy, a natural candidate for $\boldsymbol{I}$ would be to include all observed stocks they believed had any effect on production to ensure $\boldsymbol{X} \subseteq \boldsymbol{I}$. Therefore, a goal may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg 1999). However, in the interim, limits on institutional capacity for the scope and precision of measurement will likely cause the set of measured inputs $\boldsymbol{I}$ to differ from $\boldsymbol{X}$ and cause some error when estimating the growth rates and output elasticities of factor inputs.[5] In the interim, the set of measured inputs $\boldsymbol{I}$ available to practitioners remains limited. Statistical agencies often label their versions of the estimator in (4) *multifactor* productivity (as opposed to *total factor* productivity) to acknowledge that they measure productivity growth beyond the factors they are able to account for (Organisation for Economic Co-operation and Development 2001).[6]

Environmental inputs are a leading example of factors thought to affect production that are currently excluded from residual-based approaches to productivity measurement (Meyer and Harper 2005; Feenstra, Inklaar, and Timmer 2015; European Commission and Eurostat 2025;

5. It is also possible that there are natural capital stocks that produce services necessary for output that current science is not aware of.

6. The OECD manual on productivity measurement actually states this explicitly when defining multifactor productivity: "... [t]his manual uses the MFP [multifactor productivity] acronym to signal a certain modesty with respect to the capacity of capturing all factors' contribution to output growth" (Organisation for Economic Co-operation and Development 2001).

United Nations 2025).[7] Their omission often reflects limited state capacity and the difficulty of measuring how natural capital is used in production. The factor inputs in equation (1) are flows of *services* put toward production (e.g., the number of cycles by a turbine, total hours worked by the labor force, or the amount of crude oil pumped from the ground). These flows are distinct from the stocks that create them; idle machinery, unemployed persons, and dormant subsurface minerals are not counted as inputs to production. However, in many cases measuring the totality of service flows entering production from a given resource is difficult. For example, the total harvest from a fishery in a given year may account for the harvest as an input in the fishing industry, but it will not capture the service provided by the remaining stock of fish whose reproduction helps replenish the fishery going into the next year (Hannesson 2007). Total crude oil production will capture the input of extracted petroleum, but will fail to measure how a lower level of reserves reduces underground pressurization and increases future extraction costs (Anderson, Kellogg, and Salant 2018). Other environmental inputs are difficult to quantify because they are non-market (e.g., precipitation) or public (e.g., climate) in nature and are not paid the factor income they produce.[8]

2 Testing for Omitted Inputs

To illustrate the effects of omitting natural capital inputs when estimating TFP, consider an economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, such that $\boldsymbol{X} = \{K(t), L(t), N_1(t), N_2(t)\}$. For this economy, if a national accountant includes only produced capital and labor as inputs $\boldsymbol{I}$ when estimating TFP, conditional on her estimates of factor input growth rates and output elasticities $(\hat{\boldsymbol{g}}, \hat{\boldsymbol{\gamma}})$ the Solow residual can be decomposed as

$$g_A(t)|\boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x\in\{N_1,N_2\}} \gamma_x g_x(t)}_{\text{Omitted inputs}} + \underbrace{\sum_{i\in\{K,L\}} \left[\gamma_i g_i(t) - \hat{\gamma}_i \hat{g}_i(t)\right]}_{\text{Measurement error}}. \tag{5}$$

When natural capital inputs are omitted and $\boldsymbol{X} \setminus \boldsymbol{I} \neq \emptyset$, (5) will in general fail to be an unbiased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed

7. At best, some TFP growth estimates adjust for payments to environmental factors that produce marketed goods and services where factor income is observed in government accounts. A unified framework would have to account for non-market services, such as the absorption of man-made pollution, when accounting for the totality of inputs into producing all goods and services humanity consumes.

8. In practice, when environmental services are included in estimates of productivity, changes in the flows of unmeasured services are imputed using changes in the capital stocks which produce them (Brandt, Schreyer, and Zipperer 2017; Manning and Barbier 2025). Growth across all factor inputs can then be proxied by the change in how all stocks of resources—natural, human, or produced—are used over time. However, this imputation may paint a misleading picture if changes in environmental inputs are a poor proxy for changes in the flows of services they produce. Imputation is further complicated by the fact that the levels of stocks typically influence market or shadow prices of services entering production: proven reserve levels affect the market price of Brent crude; the price of landed fish depends on how scarce the remaining stock is.

to TFP because they are spuriously absorbed by the Solow residual.[9] In turn, the Solow residuals will be correlated with changes in the omitted inputs. Correlation of the Solow residual in (5) with changes in a given natural capital stock (or other potentially-omitted input) over time is then preliminary evidence that the stock in question may be an omitted input in production. Conversely, a precise null of zero correlation between changes in a natural capital service and estimated TFP growth suggests that it does not enter production and omitting the candidate stock for inclusion is unlikely to bias measures of TFP growth.

The decomposition in (5) shows three channels by which residual-based estimates of TFP growth can be correlated with natural capital: (1) correlation between natural capital use and unobservable growth in TFP, (2) correlation with omitted inputs that affect output (in this case, the natural capital stocks which are elements of $\boldsymbol{X}$ and omitted when estimating TFP growth), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While an econometrician cannot distinguish among these three potential sources of correlation, she can test whether the Solow residuals are uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with measured TFP growth, then we cannot rule out channel (2). Finding this form of empirical correlation in the data suggests that natural capital is more likely to play a role in output.

2.1 A Univariate Example

In a world with perfect measurement of the growth rates of environmental services, we could (partially) falsify that a given natural service is an omitted input by testing whether it correlates with the Solow residuals in (5). In reality, the scope of measured environmental services is limited. Even if we were able to separately observe output and TFP growth, we are restricted to testing for correlations of residualized output with *measured* natural capital services. For concreteness, consider a case of the four input example economy above where the renewable natural capital input, N_2, stems solely from fisheries. This service encompasses the biomass fisheries provide for commercial and recreational fishing operations as well as food web regulation, nutrient cycling, and serving as a pump for underwater substrates (Holmlund and Hammer 1999). Suppose a Solow residual for this economy is constructed on an annual basis by subtracting year-over-year growth in (elasticity-weighted) capital, labor, and non-renewable resources from growth in output. The discrete time analog of (5) becomes

$$g_{A,t} = g_{TFP,t} + \gamma_2 g_{N_{2},t} \ + \sum_{i \in \{K,L,N_1\}} [\gamma_i g_{i,t} \ - \ \hat{\gamma}_i \hat{g}_{i,t}] \tag{6}$$

First, there is potential for a type one error where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute it to N_i being an omitted input in production instead of some combination of channels 1 (correlation with unobservable TFP growth),

9. It is possible that the sum of the omitted input and measurement error terms are zero in expectation and the estimator is unbiased even when inputs are omitted. We view this case as unlikely.

2 (correlation with other omitted inputs), or 3 (correlation with errors in measurement). There is also the potential for a type two error: when g_{N_i} is an omitted variable in the growth accounting equation and correlated with other omitted inputs, measurement error, and/or unobservable TFP growth, these correlations may offset and lead the econometrician correctly to fail to reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute this to natural capital playing no role in production.

The difficulty facing an econometrician seeking to rule out fisheries as an omitted input is that the ecosystem services beyond harvested biomass may go unmeasured; an estimate of $g_{N_{2,t}}$ is currently not readily available from national accounts data. While we cannot directly measure all ecosystem services fisheries provide each year, the stock of fish may be measured directly at an annual frequency. We therefore propose a short regression approach: regress the (demeaned) Solow residuals on year-over-year growth in the measured natural capital stocks, which we believe correlate with the services they produce. The univariate regression case for fish stocks, denoted $g_{S_{2,t}}$, is

$$g_{A_t} = \beta_2 g_{S_{2,t}} + \varepsilon_t. \tag{7}$$

When the portion of the Solow residuals comprised of measurement error and underlying TFP growth is conditionally uncorrelated with changes in the fish stock, the estimand β_2 identifies

$$\beta_2 = \gamma_2 \frac{\mathrm{Cov}\left(g_{N_{2,t}}, g_{S_{2,t}}\right)}{\mathrm{Var}(g_{S_{2,t}})}. \tag{8}$$

So long as changes in fish stocks are a valid instrument for changes in the services they provide (i.e., $\mathrm{Cov}(g_{N_{2,t}}, g_{S_{2,t}}) \neq 0$ and $\mathrm{Cov}(g_{S_{2,t}}, \varepsilon_t) = 0$), the output elasticity of services from fisheries is identified up to a multiplicative constant.[10] More generally, when a natural capital stock is a valid instrument for the unobserved natural service flows it produces, we can falsify whether the unobserved services are correlated with measured TFP.

2.2 Identification with Multiple Instruments

To take the theory to data, we extend it to the case with multiple stocks of natural capital acting as instruments. To fix ideas, consider that we have data on annual changes for six natural capital stocks and measured total factor productivity for a panel of 117 countries. Measured TFP growth is take from the Penn World Tables (PWT, Feenstra, Inklaar, and Timmer 2015) where it is constructed as the output growth unexplained by growth in capital and labor inputs, weighted by their measured factor shares.[11]

Let the measured stocks, $\boldsymbol{S}$, of natural capital serve as instruments for an unknown number of factors $\boldsymbol{X} \setminus \boldsymbol{I}$ that affect output but are omitted from the TFP measured in the PWT. Through

10. For this estimand to be unconditionally well-defined we also require $(g_{N_{2,t}}, g_{S_{2,t}})$ to be jointly covariance stationary.

11. See equations C4 and C5 in Appendix C of Feenstra, Inklaar, and Timmer (2015) for details.

the lens of our framework, the set of included inputs is $\boldsymbol{I} = \{K, L\}$ and the data-generating process for TFP in the PWT can be decomposed as

$$g_{A_{j,t}} = g_{TFP_{j,t}} + \sum_{x \in \boldsymbol{X} \setminus \boldsymbol{I}} \gamma_x g_{i,j,t} \ + \sum_{i \in \{K,L\}} [\gamma_i g_{i,j,t} \ - \ \hat{\gamma}_i \hat{g}_{i,j,t}]. \tag{9}$$

Consider specifications of the form

$$g_{A_{j,t}} = \sum_{s \in \boldsymbol{S}} \beta_s g_{S_{s,j,t}} + \boldsymbol{\theta}' \boldsymbol{Z}_{j,t} + \varepsilon_{j,t} \tag{10}$$

where the outcome $g_{A_{j,t}}$ is TFP growth in country j in year t taken from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015), $S_{s,j,t}$ are measurements of changes in natural capital stocks, $\boldsymbol{Z}_{j,t}$ is a vector of controls, and $\varepsilon_{j,t}$ are idiosyncratic error terms. In this multivariate case, each natural capital stock $S_{s,j,t}$ included in $\boldsymbol{S}$ is intended to identify an underlying natural capital input to production that goes potentially unmeasured. When the stocks are valid instruments (uncorrelated with measurement error and underlying TFP growth), the multivariate analog of (8) is

$$\beta_s = \sum_{x \in \boldsymbol{X} \setminus \boldsymbol{I}} \gamma_x \frac{\mathrm{Cov}\left(g_{x,j,t}, \tilde{g}_{S_{s,j,t}}\right)}{\mathrm{Var}(\tilde{g}_{S_{s,j,t}})}, \tag{11}$$

where overhead tildes denote residualized values. In the case with multiple instruments and omitted environmental inputs, individual regression coefficients β_s identify a weighted sum of output elasticities associated with the omitted capital stocks where some weights may be negative. Online appendix 2.4 verifies OLS estimates of β_s are consistent and asymptotically normal under standard assumptions when the stocks $\boldsymbol{S}$ are valid instruments for the omitted inputs.

2.3 Data on Natural Capital

We examine the relationship between measured TFP growth and the environment using six country-level natural capital stocks measured in the 2024 version of the World Bank's The Changing Wealth of Nations (CWON) dataset. Columns one and two of table 1 show the six renewable capital stocks in the CWON that form the set of environmental inputs we use as covariates along with the native units of measurement for quantities over time. The timber land variable is constructed based on the total hectares of productive timber area at the country level as measured by the Food and Agriculture Organization of the United Nations (FAO). The agricultural land variable is constructed using data from the World Bank's World Development Indicators series based on total crop and pasture area measured by the FAO (World Bank 2024a). Non-timber forest cover measures the total area of forests providing ecosystem services beyond wood products and is intended to capture the change in "... the broader ecological contributions of forests, including carbon storage, water regulation, and the provision of non-wood forest products (NWFP)" over time (World Bank 2024a). CWON data on mangrove ecosystem coverage are taken from the Global Mangrove Watch (1996 -

2020) Version 3.0 Dataset (Bunting et al. 2022). Data on annual fisheries biomass were provided to the authors of the CWON by researchers at the Sea Around Us research initiative at the University of British Columbia (Pauly, Zeller, and Palomares 2020). Finally, the CWON data on annual hydroelectric power generation are taken from Midsummer Analytics (World Bank 2025).

Column three of Table 1 gives the unmeasured environmental services that we hypothesize are identified by changes in each renewable capital stock. For example, agricultural land provides nutrients for crops and grazing space for animal agriculture, non-timber forests provide recreation, mangroves protect produced capital from floods, and hydroelectric power provides electricity. Increases in the quantities of these natural capital *stocks* in column one are likely associated with corresponding increases in these services. This connection highlights the practical need to use stocks as proxies for services; while resource extraction such as timber harvest and total landed catch are often tracked by statistical agencies, they are omitted as inputs from existing calculations of TFP (Feenstra, Inklaar, and Timmer 2015). Many other ecosystem services that are recognized to affect production (e.g., water purification from forests, flood prevention from mangroves, and nutrient cycling by fisheries) go unmeasured in economic statistical data entirely in many countries, rendering their inclusion in TFP calculations infeasible.

Column four of Table 1 gives the potential confounders associated with the natural capital stocks used as instruments. While in a perfect world the stocks would only correlate with the omitted services we hope to identify in Column 3, they may also be correlated in the opposite direction with additional services that play a role in production. As shown in equation (11), β_s is in general a non-convex combination of the output elasticities associated with omitted factor inputs. When the covariances $\mathrm{Cov}(g_{s,j,t}, g_{x,j,t})$ do not all (weakly) share the same sign across the underlying omitted inputs, β_S can be zero even when the set of omitted variables contains inputs for which S_s is a good instrument. The services listed in column 4 are those we believe play a role in production while having a negative correlation with the associated natural capital stock. For example, an increased fish or timber harvest decreases the level of ecosystem services that are no longer produced by the extracted biomass. Increasing the scope of protected forests or the coverage of coastal mangroves may reduce the available land for animal grazing or marine recreation.

As the CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2019, columns five and six show the coverage of each of the six natural capital stocks in terms of total missing entries as well as the number of countries covered (out of the 117 we consider). In general, coverage of land assets is nearly complete, while coverage of mangroves, fisheries, and hydroelectric generation is sparse. When taking these data to the regression model, we impute missing values as zero changes from the last period where measurements were available, leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020. Sections 2.2 and 2.3 of the online appendix give a discussion of the imputation process for missing data along with robustness checks.

Our baseline empirical specification is a two-way fixed effects model regressing $g_{A_{j,t}}$ on growth in the six natural capital stocks ($g_{S_{n,j,t}}$) in Table 1. Equation (12) shows the empirical analog of

(5) that we take to the available data. We estimate (12) on a balanced panel of 117 countries over the 1996-2019 period:

$$g_{A_{j,t}} = \sum_{s \in \boldsymbol{S}} \beta_s g_{S_{s,j,t}} + \theta' \boldsymbol{Z}_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t}, \tag{12}$$

where j and t index countries and years respectively and the set $\boldsymbol{S}$ covers the six natural capital stocks in the CWON. The vector $\boldsymbol{Z}_{j,t}$ allows for country-level controls for produced capital, employment, labor's share of income, and human capital as well as idiosyncratic time trends. Terms (δ_j, ζ_t) allow for country and year fixed effects, and $\varepsilon_{j,t}$ capture residual idiosyncratic variation in measured productivity growth.

2.4 Inference

The β_s terms in equation (11) are our estimands of interest. While the sign on these coefficients is ambiguous and depends on the weighted summations over correlations with unobservable service inputs, our prior is that the coefficients for these six natural capital stocks will be positive. For instance, extraction from fisheries and timber resources or larger amounts of power produced by hydroelectric generation should result in greater output. A larger area of land available for economic activity allows for more structures, agriculture, recreation, and inhabitants. Thus, we expect an expansion in any of these natural capital stocks to be positively correlated in growth with output and in turn TFP when they are omitted from its construction. However, the correlation between natural capital inputs and TFP need not be positive generally; they will instead depend on physical measurement. Changes in environmental services such as heat during the summer and floods may instead be negatively correlated with output as they capture degradation in the underlying natural capital (Lemoine, Hausman, and Shrader 2026).

The joint null hypothesis of interest is that the vector $\boldsymbol{\beta} \triangleq [\beta_1, \ldots, \beta_{|\boldsymbol{S}|}]$ of coefficients on all natural capital stocks is equal to zero:

$$H_0 : \boldsymbol{\beta} = \mathbf{0}. \tag{13}$$

When the natural capital stocks are valid instruments, failing to reject this null indicates that we cannot reject the services they provide are uncorrelated with measured TFP growth. This is distinct from a causal interpretation of any one coefficient; rejecting this null does not imply changes in individual natural capital stocks cause productivity growth. Rather, rejecting the null here is suggestive evidence that it is difficult to rule out that the instrumented environmental services play no role in the production of market goods and services.

Three assumptions must hold in order for natural capital stocks to be valid instruments and to interpret the rejection of the null hypothesis in (13) as evidence that natural capital inputs are omitted when calculating TFP. First, measurement error in the (weighted) factor input growth terms used to construct estimates of TFP must be uncorrelated with measured growth in natural capital stocks. Second, natural capital stocks must be good proxies for changes in the flow of services they produce; they are strongly correlated with the services they are intended to instrument and uncorrelated with any other omitted factor inputs beyond the natural capital stocks we consider. Third, the rate of true technical change g_{TFP} is conditionally uncorrelated with changes in the stock of environmental capital.[12] These assumptions are strong. However, their failure would not indicate that natural capital is irrelevant for national accountants. Rather, if the first assumption fails, then better measurement of natural capital improves the measurement of factor inputs that currently enter national accounts even if measurement of natural capital provides no direct value when calculating productivity growth. If the second assumption fails, then it is a powerful argument for adjusting the natural capital stocks we measure in order to keep better track of the aspects of the environment that are important for the economy. Finally, if the third assumption fails, then measurement of natural capital is a first-order priority for national accountants in order to determine how nations may adjust the use of the environment to accelerate true productivity growth.

2.5 Results

Table 2 displays estimates $\hat{\beta}_s$ for each natural capital stock growth term across specifications of (12). The specification in Column (1) displays the results from a multivariate version of (7) where measured TFP growth is regressed only onto growth in the six natural capital stocks we consider. Columns (2) through (5) progressively add controls for other factor inputs in the PWT, two-way fixed effects, and idiosyncratic trend terms at the country level. Column (6) adds controls for two years of lagged values of TFP growth at the country level to account for serial correlation and is estimated using the Arellano and Bond (1991) dynamic panel estimator to avoid Nickell bias.[13]

The pair of rows in the third panel in Table 2 show p-values from two tests of the null hypothesis in (13) (i.e., that the vector of coefficients on growth in natural capital $\boldsymbol{\beta}$ is the zero vector). The bottom rows of table 2 show p-values from two Wald tests of this joint null hypothesis, one using asymptotic (robust) standard errors and a second using a semi-parametric wild bootstrap approach clustered at the country level (Roodman et al. 2019). We can reject that the natural capital stocks we consider are uncorrelated with productivity growth at traditional significance levels (5 percent) under the tests using asymptotic standard errors across five of six specifications. The tests using the wild bootstrap approach hover around the 5 percent threshold across all specifications we consider. Almost all point estimates are positive, the expected sign if the natural capital stocks we consider are strong instruments for beneficial inputs for production, although imprecisely estimated.[14]

12. Several features of measurement may strain these assumptions in practice. First, estimating output elasticities is nontrivial even for cases where factor income can be reasonably attributed (Caselli and Feyrer 2007). Second, the lack of attributed factor income makes it difficult to delineate which changes in measured natural capital stocks constitute proxies for changes in flows entering production. Finally, while measures of some forms of innovation exist (e.g., patent filings and seed varietals, used as proxies for innovation in Acemoglu et al. (2023) and Moscona and Sastry (2023) respectively), underlying correlations between environmental change and TFP growth are not falsifiable.

13. In practice, the coefficients on lagged growth terms prove to be quite significant—the p-value on a test of joint nullity is less than 0.001—consistent with documented evidence of autocorrelation in productivity growth. However, we can reject the presence of unit roots in the TFP growth panel using a Levin-Lin-Chu test at traditional levels.

14. The negative coefficient on forest cover may be driven by the confounders we speak to in Table 2. When increases in forest cover coincide with declines in logging or forestry activities that are counted in final output, the

Robustness: The results in Table 2 hold under systematic approaches to model selection as opposed to the additive approach we deploy in the main text. When (12) is estimated using LASSO-based model selection methods proposed in Belloni, Chernozhukov, and Hansen (2014), p-values from wild-bootstrap tests of the joint null hypothesis in (13) are significant at the 1 percent level across all three available procedures for covariate selection. The benchmark post-double-selection approach produces p-values of 0.002 and 0.004 for the asymptotic and semi-parametric tests of the joint null, respectively. When we implement specifications (4) and (5) using Driscoll and Kraay 1998 standard errors allowing for cross-sectional and two years of serial dependence, p-values for joint tests lie well below the 1 percent level (see Table 3 in the online appendix). A randomization-inference test of the joint null hypothesis under the two-way fixed effects specification in column 3, which tests the frequency at which the F-statistic from a Wald test of the joint null is larger than the value produced by the observed data when resampled under a true null hypothesis, produces a p-value of 0.019 (see Section 2.3.1 of the online appendix).

Individual tests of coefficient precision give some evidence that the services from land (especially those from agricultural land) are the inputs most likely to affect measured TFP. To ensure missing land rents are not the sole drivers of the results here, we re-estimate equation (12) after excluding agricultural and timber land from the set of natural capital stocks we include as covariates. Under the no-land specification, our estimates lose precision and we are only able to reject the null hypothesis at the 10 percent level (see Table 4 in the online appendix). This confirms the suggestion in Feenstra, Inklaar, and Timmer (2015) that better accounting for land rents in national accounts is a fruitful direction for future work. Online appendix 2.3.3 shows that results hold when using alternative measures of productivity growth produced by government statistical agencies as outcome variables. Tables 5 and 6 show results hold when we add additional natural capital stocks in the form of non-renewable resources and urban land as covariates when estimating (12); table 7 shows the results are also present when we instead use an alternative measure of multifactor productivity for a panel of countries in the EU plus Norway taken from Eurostat. Appendix Tables 10 and 11 test the joint null of no correlation between changes in natural capital stocks and growth in real and PPP-adjusted output at the national level and after controls are included to partial out the effects of produced capital, labor, and other factor inputs.

3 The Marginal Inclusion of Potentially Omitted Inputs

Section 2 provides suggestive evidence that natural capital is an omitted input in existing measures of TFP. We now turn to examining whether adjusting these measures to account for previously omitted inputs improves measurement. We assume the sequence of TFP growth terms, $\{g_{TFP}(t)\}$, is independent of growth in factor inputs and that all measurement of input growth is done without

correlation of increased forest cover with output may (on net) be negative. This does not per se imply that forests are a drag on the economy, but is rather indicative that the services they produce (e.g., recreation or water filtration) are not captured by measured output. Generally, any societal benefit produced by a given natural capital stock that is conditionally uncorrelated with market output will not cause a correlation with TFP growth in this exercise.

error. In this case, if a national accountant knew the true set of factor inputs to measure, the Solow residual in (4) is an unbiased estimator for $g_{TFP}(t)$ when $\boldsymbol{I} = \boldsymbol{X}$ and output elasticities are known.[15]

How does the accuracy of the Solow residual as an estimator of TFP growth change when measurement of factor inputs is incomplete? Consider an economy where the set of inputs in aggregate production $\boldsymbol{X}$ comprises labor, produced capital, and at least $J \geq 1$ additional factors which we will refer to as (omitted) natural capital stocks indexed by $\{N_j(t)\}_{j=1}^{J}$. Suppose a national accountant measures growth in output, produced capital, and labor inputs. Setting $\boldsymbol{I} = \{L(t), K(t)\}$, with estimated output elasticities for produced capital and labor ($\{\hat{\gamma}_K, \hat{\gamma}_L\}$) in hand she can construct a baseline version of the Solow residual:

$$\hat{g}_A(t) \triangleq g_Y(t) - \hat{\gamma}_K g_K(t) - \hat{\gamma}_L g_L(t) \tag{Model 1}$$

that accounts for output growth unexplained by produced capital and labor. We'll refer to this estimator as Model 1. Let $\varepsilon \triangleq \hat{g}_A - g_{TFP}$ as shorthand for the difference between the Solow residual and underlying TFP growth. When factor growth is measured without error, the bias of the estimator in Model 1 conditional on estimated output elasticities is:[16]

$$\mathbb{E}[\varepsilon|\{\hat{\gamma}_K, \hat{\gamma}_L\}] = [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}]. \tag{14}$$

Let MSE(1) denote the mean squared error (MSE) for Model 1.

Now, suppose the national accountant is also able to measure the use of the first natural capital input and its growth over time (g_{N_1}). She estimates a separate set of output elasticities associated with the new set of inputs to aggregate production, $\{\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_1\}$. Her alternative Solow residual that accounts for the first natural capital input is:

$$\tilde{g}_A(t) \triangleq g_Y(t) - \tilde{\gamma}_K g_K - \tilde{\gamma}_L g_L(t) - \tilde{\gamma}_1 g_{N_1}(t) \tag{Model 2}$$

We refer to this alternative estimator that is inclusive of the first natural capital stock as Model 2. Letting $\delta \triangleq \tilde{g}_A(t) - g_{TFP}(t)$, the conditional bias for Model 2 is

$$\mathbb{E}[\delta|\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_1] = \mathbb{E}[\gamma_K - \tilde{\gamma}_K]\mathbb{E}[g_K] + \mathbb{E}[\gamma_L - \tilde{\gamma}_L]\mathbb{E}[g_L] + \mathbb{E}[\gamma_1 - \tilde{\gamma}_1]\mathbb{E}[g_{N_1}] + \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j}]. \tag{15}$$

Let MSE(2) denote the mean squared error for Model 2. Thus the marginal inclusion of natural capital when estimating TFP growth using residual-based methods amounts to switching from Model 1 to Model 2. The biases and mean squared errors for each estimator depend on estimated

15. We additionally assume throughout that the data-generating processes for growth in inputs and TFP are stationary and that the relevant moments for these processes exist such that all estimands are well-defined.

16. Going from Model 1 to (14) implicitly assumes that the expectations of $\bar{g}_L$ and $\bar{g}_K$ are mean-independent of estimated output elasticities. This assumption, unlikely to hold in practice, is unnecessary for the results below and is used only to avoid the additional notation required for conditioning expectations.

and underlying output elasticities and the expected growth rates of inputs.

We now show when the bias of TFP growth estimates can be signed, and examine conditions under which including an omitted input on the margin reduces bias. Our first result is formed when the production function displays constant returns to scale (CRS) and factor markets are competitive. Under these assumptions, the shares of national income accruing to factors are equal to their output elasticities when the production function is log-linear. The income-based approach is the workhorse method for estimating factors' marginal products in aggregate production (Caselli and Feyrer 2007; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis 2019). Solow residuals generated by Model 1 then have the following properties.

Proposition 1. When production exhibits CRS, if $\min_j\{\mathbb{E}[g_{N_j}]\} > \max\{\mathbb{E}[g_K, g_L]\}$, then $\mathbb{E}[\varepsilon] > 0$ and Model 1 overestimates TFP growth. Conversely, if $\max_j\{\mathbb{E}[g_{N_j}]\} < \min\{\mathbb{E}[g_K, g_L]\}$, then $\mathbb{E}[\varepsilon] < 0$ and Model 1 underestimates TFP growth. In either case, going from Model 1 to Model 2 unambiguously reduces the magnitude of bias.

Proof. See Section 6.1. □

If average growth in included factors is much greater or much less than growth in all omitted factors, the bias of the Solow residuals can be signed. When all of the omitted inputs grow at a rate that is larger than the two included factors, shifting their weights γ_j in the growth accounting equation onto factors that grow slower on average leads to an underestimation of factor input growth and an upward bias in estimated TFP. The converse holds for the opposite case where all omitted factors grow more slowly than the included ones. In addition, in this case including any of the omitted inputs on the margin reduces the magnitude of the bias in estimated TFP, as it pushes the average growth of included inputs closer to the weighted average across all inputs. However, given the measurement of growth in the factor inputs that are omitted is likely to go unobserved, the conditions are difficult to test in practice. When the conditions in Proposition 1 fail to hold, we get:

Corollary 1. When growth in factor inputs does not meet either condition in Proposition 1, unless $J = 1$, going from Model 1 to Model 2 has ambiguous effects on the magnitude of bias in estimated TFP growth. The ambiguity remains even if output elasticities for all included factors are measured without error.

Proof. See Section 6.2. □

Unless going from Model 1 to Model 2 leads to no remaining omitted inputs in the Solow residual (i.e., when $J = 1$ in the case where Model 1 includes only produced capital and labor), it is ambiguous whether accounting for omitted inputs on the margin reduces the absolute level of bias when estimating TFP. Thus in the broadest sense, the additional measurement of factor inputs and apportioning of income they generate does not guarantee improved measurement of productivity growth.

The implications of the results above for the case of including non-renewable resource rents when calculating TFP growth empirically prove illustrative. Growth accounting research has long recognized that two-factor models of production are inappropriate for oil-producing economies (e.g., Mankiw, Romer, and Weil 1992). To account for this, a national accountant may adjust her residual-based TFP estimate by going from a baseline model inclusive only of produced capital and labor (Model 1) to one that also includes growth in the use of non-renewable resources, indexed N_1 henceforth, that resembles Model 2. In research settings, this has been put into practice by splitting the factor income apportioned to produced capital under Model 1 between produced and natural capital (e.g., Caselli and Feyrer 2007; Brandt, Schreyer, and Zipperer 2017; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis 2019; Manning and Barbier 2025). Under the assumption of CRS production and competitive factor markets, an estimate inclusive of natural capital can be constructed by setting $\tilde{\gamma}_1$ to the share of national income accruing to privately-owned non-renewable resources and then recalculating the output elasticity of produced capital as one less than the income shares for labor and natural capital, $\tilde{\gamma}_K = 1 - \tilde{\gamma}_L - \tilde{\gamma}_1$. The growth term for non-renewable resources $g_{N_1}(t)$ is then constructed as the change in the extraction of physical resources over time.

Corollary 1 implies that this process of apportioning capital income between produced and natural capital can increase the magnitude of the bias in measured TFP. Even in the extreme cases where all elasticities γ for the included factors are known without error, unless the conditions in Proposition 1 hold, including measurement of non-renewables on the margin does not necessarily lower the magnitude of bias in estimated TFP relative to Model 1. The negative result above translates into the other measure of goodness of fit we consider, mean squared error:

Proposition 2. When $J \geq 2$ it is ambiguous whether MSE(2) is lower than MSE(1), even if output elasticities are measured without error.

Proof. See Section 6.3. □

Proposition 2 presents the non-monotonicity result that motivates this paper. Only when all growth terms (weakly) share the same sign in expectation and all output elasticities are measured without error is it guaranteed that MSE(2) is no larger than MSE(1). This holds for generalized versions of Model 1 and Model 2 (e.g., when Model 2 includes more than one previously-omitted input on the margin) unless going from Model 1 to Model 2 makes it so the Solow residual is correctly specified.

An alternative formulation for the bias in these estimators is to express the omitted variables as deviations from average growth across all natural capital factors that are omitted from a given model. Let $\boldsymbol{J} \subset \boldsymbol{X}$ be the set of omitted factors under Model 1 and $\boldsymbol{J}' \subset \boldsymbol{J}$ be the set of factors omitted under Model 2. The average output elasticities and growth rates across these factors are

$$\bar{\gamma}^{\boldsymbol{J}} \triangleq \frac{1}{|\boldsymbol{J}|} \sum_{j \in \boldsymbol{J}} \gamma_j \quad , \quad \bar{g}^{\boldsymbol{J}} \triangleq \frac{1}{|\boldsymbol{J}|} \sum_{j \in \boldsymbol{J}} \mathbb{E}[g_{N_j}]$$

such that the bias terms can be written as

$$\mathbb{E}[\text{Bias}|\boldsymbol{J}] = |\boldsymbol{J}|\Big(\text{Cov}\Big(\gamma_j, \mathbb{E}[g_j]\Big|j \in \boldsymbol{J}\Big) + \bar{\gamma}^{\boldsymbol{J}}\bar{g}^{\boldsymbol{J}}\Big) \tag{16}$$

when output elasticities for included inputs are measured without error. When the expected growth rates of natural capital inputs are uncorrelated with output elasticities for the set of omitted inputs, the covariance terms above become zero and a sufficient condition for declining absolute bias becomes

$$\frac{|\bar{\gamma}^{\boldsymbol{J}}\bar{g}^{\boldsymbol{J}}|}{|\bar{\gamma}^{\boldsymbol{J}'}\bar{g}^{\boldsymbol{J}'}|} \geq \frac{|\boldsymbol{J}'|}{|\boldsymbol{J}|} \tag{17}$$

The converse to Proposition 2 is that there is always a possibility for better measurement to lower bias and MSE. All combinations of additional measurement—going from omitting $\boldsymbol{J}$ to $\boldsymbol{J}'$—where including additional inputs reduces the product of the average growth and output elasticity of omitted factors (in absolute terms) lowers bias and MSE when input growth is independent (see the proof in 6.3 for details).

The result that the marginal inclusion of natural capital (or any omitted capital stock) need not reduce bias in estimates of TFP growth has, to our knowledge, gone unrecognized in the growth accounting literature. For example, existing work by Brandt, Schreyer, and Zipperer (2017) and Monge-Naranjo, Sánchez, and Santaeulalia-Llopis (2019) and Manning and Barbier (2025) examines how the inclusion of natural capital inputs affects income-based growth accounting estimates of TFP. The results here show it is not possible to know whether these adjustments for non-renewable resources bring the resulting alternative estimates closer to the rate of true productivity growth over time unless one assumes that they render the growth accounting equation correctly specified.

4 Simulating the Effects of Incorporating Natural Capital when Estimating TFP

Section 3 examined the theoretical effects of reducing the number of omitted factor inputs on the performance of the Solow residual as an estimator of TFP. This section evaluates this question using the available data on growth in factor inputs and natural capital stocks to take a version of the four-input example economy from Section 2 to the data. Drawing on empirical moments of factor input growth and output elasticities from the literature and our own estimates, we calibrate data-generating processes for output across a panel of countries in the CWON. We examine whether the marginal inclusion of an omitted factor input improves accuracy when TFP growth is estimated on these simulated data.

4.1 Setup

We consider a case where a national accountant seeks to estimate TFP growth for the 100 countries in the CWON dataset with data on renewable and non-renewable natural capital stocks.[17] Output in each country is determined by TFP and four factor inputs: produced capital (K), labor (L), non-renewable natural capital (N_1), and renewable natural capital (N_2). To motivate the notion of her having to decide whether to include non-renewable capital on the margin, we assume the national accountant's measurement of factor input growth is incomplete. She measures output and factor input growth without error, but measures only growth in the use of produced capital, labor, and non-renewable capital. She suspects that services from renewable capital are important inputs in production, but is unable to measure how their use has changed over time. Knowing this makes it ambiguous whether including non-renewable resources improves her estimate of TFP growth, she must choose whether to include growth in the use of non-renewable resources when taking equation (4) to the data. In this thought experiment, a model is preferred when it has a lower root mean squared error (RMSE) when estimating average TFP growth on data for 25 years' worth of growth in output and factor inputs.[18]

Data-Generating Process for Output: Output growth in each country comprises growth in TFP, produced capital, labor, and the two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vectors $\boldsymbol{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j. Output growth in each country-year pair is generated by

$$g_{Y,j,t} = \boldsymbol{g}'_{j,t}\boldsymbol{\gamma}_j \tag{18}$$

where $\boldsymbol{g}_{j,t} \triangleq [g_{TFP,j,t}, g_{K,j,t}, g_{L,j,t}, g_{N_1,j,t}, g_{N_2,j,t}]'$ is the vector of determinants of TFP growth and $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for country j.

Calibrating Country-Level Growth: We assume at the country level that growth in TFP ($g_{TFP,j,t}$) is uncorrelated with growth in other factors. We calibrate the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\boldsymbol{g}_{j,t}$ in each country based on their analogs in the CWON and PWT. The mean and covariances of growth in produced capital services, labor, and TFP are set equal to their sample analogs in the Penn World Tables over the same 1996-2019 period.[19] For natural capital inputs, we follow the World Bank CWON methodology (World Bank 2025) and construct aggregate Törnqvist quantity indices for non-renewable and renewable resource use at the country level and set the parameters governing expected growth in the use of natural

17. The sample here shrinks from the one in Section 2 due to missing data on non-renewables.
18. We round up to 25 from the size of 24 years in our estimating sample.
19. Section 4.5.2 of the Appendix shows that re-estimating the sample moments used to calibrate country-level data generating processes based on two partitions of the original 1996-2019 data leads to larger estimates of RMSE reductions across the board and minimal qualitative changes in the set of countries for which ΔRMSE is negative.

capital and its second moments equal to the sample means of growth in these indices between 1996 and 2019 (Törnqvist et al. 1936).[20] Table 3 below gives a full accounting of all data used in the simulation exercise along with the mapping between sample and population moments.[21]

Output Elasticities: We calibrate output elasticities to be the same across countries such that $\boldsymbol{\gamma}_j = \boldsymbol{\gamma}$. Online appendix sections 4.2 and 4.3 show that among the elasticities governing output growth, only those on the two natural capital stocks affect the RMSE for the intercept terms. As such, we set the output elasticity of services from renewable natural capital, γ_2, to equal 0.05 based on the analysis in Section 3.6 of the online appendix. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al. 2004), we set the output elasticity of services from non-renewable natural capital, γ_1, to 0.05. An extended justification for this calibration is provided in online appendix 4.1. However, because this parameter choice is external to the model, we perform a sensitivity analysis of our results to these choices of elasticities for natural capital after discussing our initial results.

Procedure for Estimating TFP Growth: The national accountant observes 25 years' worth of data on growth in factor inputs K, L, and N_1 and output Y for each country. We assume she does not observe the distribution of national income earned across factors, and therefore cannot estimate output elasticities using factor shares. This ensures our thought experiment is consistent with construction of Solow residuals in empirical settings, where factor income generated by non-market natural capital services (regardless of whether they are renewable) is unobservable to an econometrician (Manning, Taylor, and Wilen 2018).

To construct a measure of average TFP growth that does not rely on growth in non-renewable resources, the national accountant estimates the following regression:

$$g_{Y,j,t} = \gamma_{A,j} + \gamma_K g_{K,j,t} + \gamma_L g_{L,j,t} + \epsilon_{j,t} \tag{19}$$

on time series of data for a given country using ordinary least squares (OLS) where $\gamma_{A,j}$ is the country-level intercept term. Dropping country subscripts, we write her estimator for average TFP growth using only produced capital and labor, the intercept term produced by estimating (19), as $\hat{\gamma}_A$. We refer to this estimator as model 1. She then estimates the analog for the case including non-renewable resources:

$$g_{Y,j,t} = \gamma_{A,j} + \gamma_K g_{K,j,t} + \gamma_L g_{L,j,t} + \gamma_1 g_{N_1,j,t} + \nu_{j,t} \tag{20}$$

on time series for growth in produced capital, labor, and non-renewables. We write the OLS estimator for average TFP growth inclusive of non-renewable capital as $\tilde{\gamma}_A$ and refer to this estimator

20. Section 3 of the online appendix outlines the full procedure for how we construct the Törnqvist indices for both forms of natural capital and calibrate $\boldsymbol{\mu}_j$ and Σ_j for each country.

21. See Section 4 of the online appendix for a full documentation of how the simulations are calibrated as well as our choice of parameters.

as model 2.

Calculating RMSE for each Model: Dropping country subscripts, we calculate the RMSE for $\hat{\gamma}_A$ and $\tilde{\gamma}_A$ against the calibrated values $\mathbb{E}[g_{TFP}]$ by evaluating models 1 and 2 at the sample analogs of $\boldsymbol{\mu}$ and Σ for a given country. The change in RMSE produced by including non-renewables on the margin—going from model 1 to model 2—is given by:

$$\Delta\text{RMSE}|\boldsymbol{\mu}, \Sigma \triangleq \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} - \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} \tag{21}$$

Equation (21) gives the expected change in the root mean squared error a national accountant's estimate of average TFP growth induced by switching from model 1 to model 2 in a given country. We repeat the process in (21) for each of the countries in the CWON with sufficient data to form a cross section of 100 country-level observations of ΔRMSE.[22]

4.2 Simulation Results

For 94 out of 100 countries we consider, including non-renewable resources (going from model 1 to model 2) lowers the RMSE of estimated TFP growth. The relative performance of model 2 relative to model 1 can be seen by an overwhelming share of countries mapped in green in Figure 1a; those with green shading are those where RMSE falls when non-renewables are included. The histogram in Figure 1b displays the binned frequency of country-level ΔRMSE across countries by the magnitude of changes; green countries in the first panel fall in the green bins to the left of $x = 0$ in panel 1b. The six observations in the blue bins to the right of $x = 0$ are countries for which including non-renewables increases the RMSE of estimated TFP growth (and decreases accuracy under this criterion).[23] The median and mean declines in RMSE are sizable: reductions of 7.3 and 32 basis points (one-hundredth of a percentage point, abbreviated as bps) respectively. Reductions in error are dispersed (the standard deviation is 66 bps) and left-skewed. The countries with the largest reductions in RMSE (largest ΔRMSE in magnitude) are Kenya, Laos, Israel, Tanzania, and Niger. The magnitude of the median reduction is a 12.5 percent of average TFP growth across all countries over the 1996-2019 period. For the six countries where including non-renewables raises RMSE, the deterioration in fit is small (less than 1.2 basis points on average and at the median). The greatest RMSE increase occurs for Chile, where RMSE rises by 4.8 bps.[24]

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in measured natural capital use and baseline levels of calibrated productivity growth. Panels 2a and 2b plot the decline in mean squared error ($-\Delta$RMSE) induced by including non-renewables when estimat-

22. Section 4.1 of the online appendix gives the full procedure for calculating the bias and variance of the estimators in models 1 and 2.

23. The six countries are Chile, Indonesia, Ecuador, Austria, Zimbabwe, and Egypt.

24. Chile is an empirical case where the two components of bias in (19), stemming from omitting two variables, happen to offset each other almost perfectly, causing the bias when only renewables are omitted in (20) to be larger in magnitude.

ing TFP growth against the calibrated values of average growth in the use of non-renewable and renewable resources (g_{N_1} and g_{N_2}) at the country level. Of the 100 countries in our sample, 68 countries exhibit positive average growth in their use of non-renewables, and 47 exhibit positive average growth in the use of renewable inputs. Importantly, panels 2a and 2b show that average growth in non-renewables is about an order of magnitude greater than growth in renewables. This proportionally larger growth in non-renewable resources is responsible for substantially more output growth than that of renewables, even when the contribution of both resources to output in levels is similar. In turn, leaving out non-renewables on the margin leaves more unexplained output in model 1 relative to model 2 where only renewables are omitted.

Panels 2c and 2d show point estimates of TFP growth from models 1 and 2 respectively against the true data-generating process (set as the observed value in the PWT). Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. The median (mean) decrease in RMSE is 10 (23) percent of baseline error when both natural capital stocks are omitted, and rises to more than 40 percent for over a quarter of the countries in the cross section of simulated data. Reduction in the bias of the estimators drives most of the changes in RMSE rather than a smaller expected variance. Bias drives 98 percent of RMSE on average when non-renewables are excluded (model 1), and 89 percent of RMSE on average when non-renewables are included (model 2).[25]

4.3 RMSE Changes in Context

The resulting decline in measurement error (as proxied by ΔRMSE) for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 7.3 basis points. If estimates of historical averages are used to forecast future growth, this translates to an annual forecast error of roughly 0.07 percentage point. Applying this median value to the United States, when this forecast error is compounded over ten years (the standard CBO budget horizon) at current levels of output, forecast cumulative revenues would deviate from projections by approximately \$300 billion.[26] This forecast error is roughly one quarter of the aggregate social cost of compliance with the U.S. Clean Air Act between 1970 and 1990, one of the broadest environmental policies enacted globally to date (U.S. Environmental Protection Agency 1997). However, the cumulative deviation for the United States would grow to \$1.2 trillion if we were to instead apply our U.S.-specific point estimate for the change in RMSE (26 bps) when including non-renewables on the margin.

Decomposition across inputs: To decompose the contribution of individual components of output growth to changes in measurement accuracy, we regress ΔRMSE on the individual components of $\boldsymbol{\mu}_j$ and Σ_j. This gives a reduced-form sense of the elements of input growth most important

25. See Section 4 of the online appendix for the derivation and decomposition of the changes in RMSE in terms of changes in bias and variance.

26. See Section 5 of the online appendix for a description of the back-of-the-envelope calculation reported here.

in determining the effects of measuring non-renewables.[27] Online appendix Table 16 shows that ΔRMSE is decreasing in the country-level growth rates of renewable and non-renewable resource use, and increasing in the correlations between growth in non-renewable resource use and those for labor and produced capital.[28] Thus if the world looks increasingly like Spain or Kyrgyzstan—nations where growth in the use of non-renewables is negatively correlated with produced capital and labor, then including non-renewable resources in national accounts is more likely to reduce the error in future estimates of TFP growth. In contrast, if the world becomes more like Chile or Indonesia, where growth in the use of non-renewables is positively correlated with produced capital and labor, then adding non-renewables to national accounts will have a more limited impact on the error in future estimates of TFP growth.[29]

Other predictors of error reduction: The changes in RMSE resulting from including non-renewables do not appear correlated with other aggregate characteristics at the country level beyond moments governing growth in factor inputs. Figure 3 plots RMSE reductions against six additional statistics at the country level measured in 2019: (log) GDP, (log) GDP per capita, human capital levels, statistical capacity, resource rents as a share of output, and natural resource depletion as a share of national income. While some relationships are statistically significant, the amount of variation in ΔRMSE explained by these factors across countries is quite small (as indicated by the relative flatness of linear and non-parametric fits). Panels 3a and 3b indicate that the magnitude of RMSE changes tends to decrease in both measures of income. The relationship is robust and fairly substantial in magnitude; a three-fold increase in GDP decreases estimation error in TFP growth by 20 basis points (double our baseline estimate). A similar relationship holds between RMSE reductions ($-\Delta$RMSE) and the level of human capital in a given country (Panel 3c), which is unsurprising given the correlation between education and output. This relationship can be explained by higher-income countries tending to have smaller and less-volatile TFP growth during this period, which lowers the variance of both estimators and thus shrinks the error reduction in absolute terms.

Panel 3d shows the relationship between RMSE improvement and statistical capacity as measured by the World Bank for 67 countries in our sample marked as developing.[30] The correlation here is a fairly precise zero, which we interpret as suggestive evidence that better measurement of natural capital may benefit countries regardless of current national accounting practices. The last two panels take an alternative approach and examine how including non-renewables in our simulations relates to two measures of the *level* of resource intensity in production at the country-

27. The exact partial derivatives can also be calculated on a country-by-country basis using the closed-form solutions for ΔRMSE in Sections 4.2 and 4.3 of the online appendix.

28. Three of these four determinants of ΔRMSE are significant at traditional (5 percent) levels when ΔRMSE is regressed on the elements of $\boldsymbol{\mu}_j$ and Σ_j that enter the RMSE calculations. The joint test is significant at the 0.1 percent level. See Section 4.4 of the online appendix for details and reduced-form analysis of how other sample moments determine ΔRMSE.

29. Figure 4 in Section 4.4 of the online appendix shows a version of these two panels for all 12 sample moments that determine ΔRMSE at the country level.

30. Statistical capacity is measured only for developing countries as defined by the World Bank.

level. Panel 3e shows RMSE reductions against resource rents as a share of GDP taken from the World Bank's World Development Indicators database (World Bank 2024b), while panel 3f plots the improvements against an alternative metric for natural capital intensity, resource depletion as a share of gross national income. Perhaps surprisingly, there is a precise negative (albeit small in magnitude) statistical relationship between measures of resource dependence and error reduction. For highly resource-dependent economies (e.g., Iraq and Kuwait), RMSE reductions are some of the smallest in our sample (1 and 3 basis points respectively). The negative relationship persists qualitatively after controlling for the other factors that have negative correlations shown here (i.e., human capital and GDP), but loses statistical significance and flips signs when conditioned on historical volatility in TFP. Thus a high reliance on resource rents (typically driven by non-renewable extraction) being associated with ΔRMSE closer to zero does not imply that measurement is less-important for resource-dependent countries, but rather that resource-dependent countries experience more volatile growth.

Comparison with existing results: The magnitudes of ΔRMSE induced by including non-renewables here are similar to the adjustments to TFP accounting for non-renewables calculated by Brandt, Schreyer, and Zipperer (2017). The authors there find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries between 1986 and 2008. Manning and Barbier (2025) find adjusting the growth accounting equation to account for natural capital increases the amount of output growth unexplained by factor inputs by 87 basis points on average for countries with negative historical TFP growth and decreases it by 65 basis points on average for countries where TFP growth has been positive. Freeman, Inklaar, and Diewert (2021) use an earlier version of the CWON dataset to adjust TFP estimates to account for subsurface minerals and fossil fuels and find omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies.

Sensitivity analysis: We perform a Monte Carlo (MC) analysis to examine the sensitivity of our results to alternative values of output elasticities for the two natural capital stocks.[31] Each individual MC draw generates an alternative pair of output elasticities for natural capital (γ_1, γ_2) by taking two iid draws from the uniform distribution $\mathcal{U}[0, 0.1]$ and recalculating ΔRMSE_j for a random country in our sample under the new parameters. We repeat this procedure 10,000,000 times, generating roughly 100,000 alternative changes of RMSE for each country. The median (mean) reduction in RMSE at the country level across our ten million alternative draws falls to 2 (28) basis points. The distribution remains quite disperse, with a standard deviation of 80 basis points. Unlike with our main analysis, 31 percent of country-level changes in RMSE induced by including non-renewables (going from model 1 to model 2) are positive as compared to 6 percent in our main simulation.

31. We MC over these two elasticities because only γ_1 and γ_2 affect ΔRMSE in practice; see online appendices 4.2 and 4.3.

Figures 4a and 4b show the cross-sectional relationship between reductions in RMSE at the country level and the randomly-drawn values of output elasticities for non-renewable and renewable natural capital, respectively. All relationships shown are fitted after winsorizing the bottom 10 percent of the Monte Carlo draws ($\Delta\text{RMSE}_j \leq -0.0085$) to avoid the leverage produced by observations in the left tail where the reductions in RMSE are exceptionally large. Solid black lines show the fitted linear relationship between changes in (expected) ΔRMSE_j and the associated output elasticity γ_1 (γ_2) holding all else constant, and dotted orange lines show local polynomial fits. The red dashed lines show cubic splines between medians across 70 bins of γ_1 (γ_2).

Figure 4a indicates that our finding that including non-renewables becomes less likely to decrease RMSE when estimating TFP as the output elasticity of non-renewable capital becomes smaller. All three fitted relationships (linear, binned median, and local polynomial) increase monotonically in γ_1 and cross the x axis only once. This suggests that including non-renewables when measuring TFP becomes likely to increase error when the output elasticity of non-renewable capital is about 0.015.[32] However, this crossing point masks the substantial dispersion in ΔRMSE conditional on resampled γ_1 values. For resampled values less than the calibrated value in the main text (i.e., those where $\gamma_1 \leq 0.05$), 50.8% of Monte Carlo trials produce increases in RMSE and including non-renewables worsens measurement at the country level.

Figure 4b displays the cross section of RMSE across Monte Carlo trials conditional on the newly-drawn values for γ_2. For renewable resources, the fitted lines suggest changes in RMSE increase monotonically as the resampled γ_2 rises. Thus our finding that measuring non-renewables decreases RMSE when estimating TFP becomes less likely to hold as the output elasticity of renewable natural capital increases relative to the values we consider. While the average and local polynomial fits do not cross the x-axis, the binned median fit crosses the x-axis at around $\gamma = 0.075$. For resampled values larger than the calibrated value in the main text (i.e., those with $\gamma_2 \geq 0.05$), 45.9% of Monte Carlo trials produce increases in RMSE (i.e., are cases where including non-renewables worsens measurement at the country level), again indicating that the fitted values mask quite a bit of variation in ΔRMSE conditional on resampled values of γ_2. All else equal, if the use of non-renewable natural capital begins to play a smaller role in production in the future or its contribution has been overestimated in the past (the use of renewables becomes more important, or its contribution here has been underestimated), it will be less likely that including non-renewables when estimating TFP will aid measurement in the future.

5 Conclusion

Fifty-four years after Nordhaus and Tobin (1973), research continues to highlight new channels by which environmental change affects society and welfare. An increasing share of this research has focused on responding to high-level calls for improved welfare measures, e.g., the U.N. Secretary General's push for measures "Beyond GDP" (United Nations Statistics Division 2026). Economists

32. Section 4.1 in the appendix describing our calibration of model parameters discusses why we believe a value this low is unlikely empirically.

have long recognized the importance of such measures, and there is a rich literature on complements to GDP that would better account for the determinants of welfare beyond market activity (Fleurbaey 2009; Jorgenson 2018).[33] This paper gives evidence that measurement of natural capital benefits governments and other economic actors irrespective of goals beyond GDP. When natural capital affects production, omitting it from national accounts biases estimates of economic growth. This supports the original hypothesis by Nordhaus and Tobin in 1973 that leaving things out ("treating as free things that are not really free") of national accounts "... gives the wrong signals for the directions of economic growth" in addition to affecting measurement of national welfare (Nordhaus and Tobin 1973). While measuring the environment will remain important for assessing the sustainability of development, regulatory compliance, and public health, we show here that systematically connecting changes in the environment to economic statistics will benefit existing measurement that is already used to guide public policy and private investment.

33. Conte, Addicott, and Millerhaller (2025) provide micro-founded support for this position regarding the environment, demonstrating how reductions in natural capital stocks can generate real income effects for consumers.

6 Proofs

6.1 Proposition 1

Proof. We calculate the bias for the estimator in Model 1 conditional on estimated output elasticities for capital and labor $\{\hat{\gamma}_K, \hat{\gamma}_L\}$. Note that with CRS production and competitive markets, estimated elasticities for the two-factor estimator must satisfy $\hat{\gamma}_K + \hat{\gamma}_L = 1$. Proposition 1 can then be proved in parts.

Part 1: Let $\bar{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\underline{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$. Suppose that $\underline{g}_N > \bar{g}$. Then

$$
\begin{aligned}
\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1-\hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&\geq \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] + [\gamma_K]\mathbb{E}[g_K] + [\gamma_L]\mathbb{E}[g_L] - \bar{g} \\
&\geq \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] + [\gamma_K + \gamma_L]\bar{g} - \bar{g} \\
&= \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] - \sum_{j=1}^{J} \gamma_j \bar{g}
\end{aligned}
$$

where the first and second inequalities follow from the definition of $\bar{g}$ and the last equality follows from $1 - \gamma_K - \gamma_L = \sum_{j=1}^{J} \gamma_j$. We then have

$$
\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] \geq [\underline{g}_N - \bar{g}] \sum_{j=1}^{J} \gamma_j \geq 0
$$

as $\underline{g}_N - \bar{g} > 0$ by assumption. Finally, note that adding a previously-omitted factor input term lowers bias, as we reduce the number of positive factor shares γ_j in the summation in the last line by increasing the starting index to $j = 2$ (without loss of generality).

Part 2: Let $\underline{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\bar{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$ and suppose that $\underline{g} > \bar{g}_N$. Then analogous to the presentation above we have

$$
\begin{aligned}
\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1-\hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&\leq \sum_{j=1}^{J} \gamma_j g_{N_j} - \sum_{j=1}^{J} \gamma_j \underline{g} \\
&\leq [\bar{g}_N - \underline{g}] \sum_{j=1}^{J} \gamma_j \leq 0
\end{aligned}
$$

where in this case adding a correctly-measured omitted input stock (removing a term from the summation) decreases bias in nominal and absolute terms.

6.2 Corollary 1

Part 1 Suppose Model 1 includes a subset of inputs $\boldsymbol{I} \subset \boldsymbol{X}$ as well as a set of associated estimated output elasticities $\hat{\boldsymbol{\gamma}}$. Assume Model 2 includes the subset $\boldsymbol{I'}$ with associated output elasticities $\tilde{\boldsymbol{\gamma}}$ where $\boldsymbol{I} \subset \boldsymbol{I'} \subset \boldsymbol{X}$. The biases in the general case are given by

$$\mathbb{E}[\varepsilon|\hat{\boldsymbol{\gamma}}] = \sum_{i\in\boldsymbol{I}}[\gamma_i - \hat{\gamma}_i]\mathbb{E}[g_i] + \sum_{j\in\boldsymbol{X}\setminus\boldsymbol{I}}\gamma_j\mathbb{E}[g_{N_j}]$$
$$\mathbb{E}[\delta|\tilde{\boldsymbol{\gamma}}] = \sum_{i\in\boldsymbol{I'}}[\gamma_i - \tilde{\gamma}_i]\mathbb{E}[g_i] + \sum_{j\in\boldsymbol{X}\setminus\boldsymbol{I'}}\gamma_j\mathbb{E}[g_{N_j}]$$

The signs and magnitudes of both right-hand terms are indeterminate without additional assumptions. Even if all growth terms are assumed to be (weakly) of the same sign in expectation, the sign of the biases will still depend on how the estimated output elasticities differ from unobserved underlying values. Even when output elasticities for the included factors are measured without error, such that both right hand sides above collapse to only the second term, it remains ambiguous whether going from Model 1 to Model 2 reduces bias.

Part 2 Suppose now that one of the output elasticities for the included growth terms is measured without error such that $\hat{\gamma}_K = \gamma_K$. Under the assumption of CRS with competitive markets, the labor share is then $(1 - \hat{\gamma}_K)$. Then the bias from the first model is:

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_L - g_{N_j}] \lesseqgtr 0$$

Alternatively, if $\hat{\gamma}_L = \gamma_L$ (the labor share is measured without error), then we have $\hat{\gamma}_K = \gamma_K + \sum_j \gamma_j$ which yields

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_K - g_{N_j}] \lesseqgtr 0$$

Part 3 Suppose instead that an econometrician is able to measure output elasticities for both included factors without error. While this causes the bias to comprise only the omitted factor terms, its sign remains ambiguous:

$$\mathbb{E}[\varepsilon|\hat{\gamma}_K, \hat{\gamma}_L] = \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_{N_j}] \lesseqgtr 0$$

□

6.3 Proposition 2

Proof. We prove the result in Proposition 2 here the result for a special case where each input growth term is iid and output elasticities and factor input growth are known to the econometrician without error. Introducing correlations in growth terms or measurement error adds more potential sources of ambiguity, but is not necessary to show that adding additional natural capital stocks has ambiguous effects on bias.

The mean squared errors of our estimators when all factor shares and growth terms are measured without error are

$$\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) = \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right]$$

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \mathbb{E}\left[\left(\sum_{j=2}^{J} \gamma_j g_{N_j}\right)^2\right]$$

Evaluating both means squared errors gives

$$\begin{aligned}
\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) &= \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right] \\
&= \text{Var}\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right) + \mathbb{E}\left[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)\right]^2 \\
&= \sum_{i=1}^{J}\sum_{j=1}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)\right]^2
\end{aligned}$$

where the last two lines substitute in the conditional bias terms from the proof of Corollary 1. For Model 2 we have the analog less the included NK stock N_1:

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \sum_{i=2}^{J}\sum_{j=2}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\delta||(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1)\right]^2$$

The difference in MSE is

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{2\gamma_1 \sum_{i=2}^{J} \gamma_j \text{Cov}\left(g_{N_1}, g_{N_j}\right)}_{\lesseqgtr 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0$$

Under the assumption of iid growth terms for all natural capital inputs g_{N_j}, the above difference becomes

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0s$$

where the inequality is ambiguous unless $\mathbb{E}[\delta] = 0$ in the case where including N_1 makes the growth accounting equation correctly specified.

$\square$

7 Tables and Figures

Table 1: Renewable natural capital stocks quantified in the Changing Wealth of Nations

Resource	**Natural Unit**	**Instrumented Service Flow**	**Potential Confounders**	**% Obs. Missing**	**# Countries**
Timber land	Hectares	Timber harvest, carbon capture	Decline in seeding services	1.25%	114
Agricultural land	Square kilometers	Grazing area, soil services	Decline in ecosystem services from fallow land	2.56%	116
Non-timber forest cover	Square kilometers	Recreation, ecosystem services	Decline in grazing area and forestry	3.95%	113
Mangrove area	Hectares	Flood protection, water filtration, embedded habitats	Marine recreation	63.25%	43
Fishery biomass	Metric tons	Fished biomass	Nutrient cycling, pump dynamics	23.08%	90
Hydroelectric generation	Gigawatt-hours	Electricity, flood prevention, water storage	Ecosystem services from rivers and floodplains	27.07%	86

Note: All data here are taken from the World Bank *Changing Wealth of Nations* database (World Bank 2025). The first two columns report each resource and the unit in which physical quantities are measured. The column labeled "Instrumented Service Flow" gives an example of services from natural capital that we believe affect TFP that are likely correlated with the change in these natural capital stocks. The "Potential Confounders" column reports how this stock may also be correlated with (1) other omitted inputs to GDP beyond natural capital or (2) measurement error for the factors that are included. The column "% obs. missing" reports the share of observations between 1996 and 2019 with missing quantities in the initial dataset. The "# Countries" column shows the number of countries where at least one observation is non-zero.

Table 2: TFP Growth vs. Growth in Renewable Natural Capital Stocks

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Timber Land	0.165	0.168	0.071	0.074	-0.004	0.025
	(0.049)	(0.046)	(0.048)	(0.046)	(0.053)	(0.104)
Agricultural Land	0.029	0.045	0.032	0.038	0.042	0.043
	(0.019)	(0.019)	(0.019)	(0.018)	(0.018)	(0.017)
Forest Cover	-0.069	-0.113	-0.006	0.010	0.081	-0.027
	(0.062)	(0.065)	(0.062)	(0.062)	(0.055)	(0.076)
Mangroves	0.112	0.057	0.352	0.322	0.318	-0.111
	(0.129)	(0.132)	(0.163)	(0.167)	(0.202)	(0.427)
Fishery Biomass	0.112	0.026	0.092	0.070	0.039	-0.109
	(0.056)	(0.054)	(0.063)	(0.061)	(0.082)	(0.105)
Hydropower	0.010	0.011	0.009	0.009	0.006	0.006
	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.001	0.001	0.018	0.018	0.090	0.045
Bootstrap Wald	0.000	0.000	0.008	0.010	–	0.062
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable at the country level, $\Delta \ln(TFP_{j,t})$, is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables enter as log changes. Robust standard errors are shown in parentheses. Rows under the "Controls for ... " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Row "Wald" displays p-values from Wald tests of the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis clustered at the country level (Roodman et al. 2019). No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 3: Simulation Moments and Data Sources

Description	**Population Moment**	**Sample Moment**	**Source**
Non-Renewable Services	$\mathbb{E}[g_{N_1}]$	$\bar{g}^{R}$	CWON
Renewable Services	$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
Produced Capital Services	$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
Labor Inputs	$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
TFP	$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Covariances	Σ	$\widehat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the covariance matrix Σ. CWON refers to the World Bank *Changing Wealth of Nations* database (World Bank 2025). PWT refers to the Penn World Tables version 10.01 (Feenstra, Inklaar, and Timmer 2015).

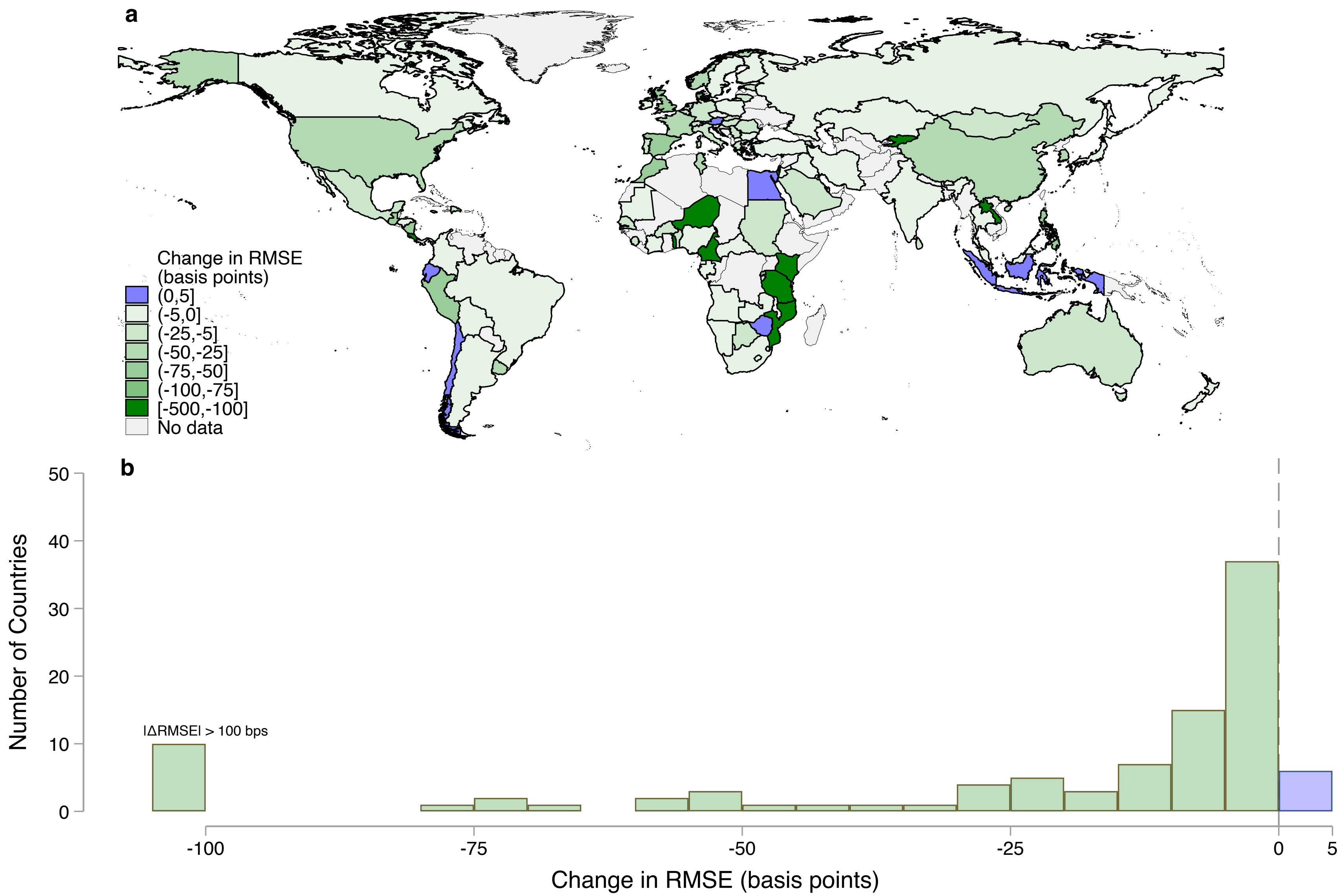


Figure 1: Panel **a**. Map of ΔRMSE for the 100 countries in our sample. Panel **b**. Histogram of the reductions in root mean squared error—ΔRMSE—when adding g_{N_1} to the Solow residual-based TFP estimator for each of 100 countries in the sample. The 10 countries with RMSE reductions larger than 100 basis points in magnitude are topcoded and added to the leftmost bin in the histogram. See Table 15 in the online appendix for a full list of country-level sample moments and changes in RMSE.

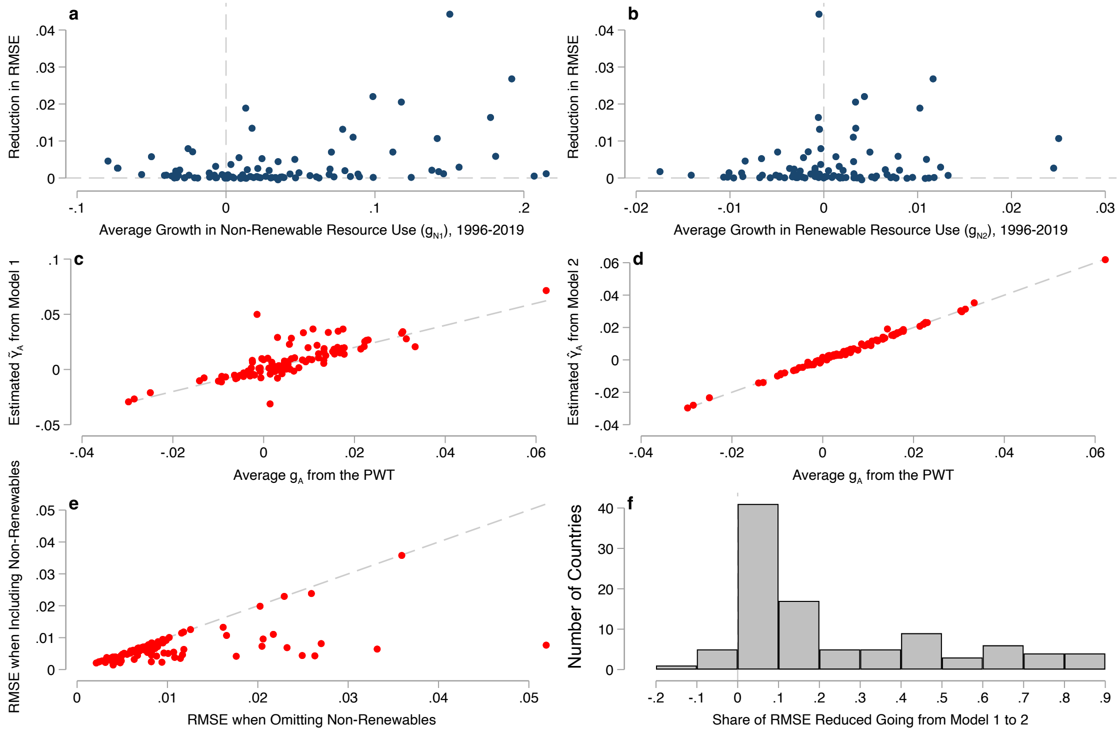


Figure 2: Panels 2a and 2b plot ΔRMSE when estimating TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data-generating process. Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

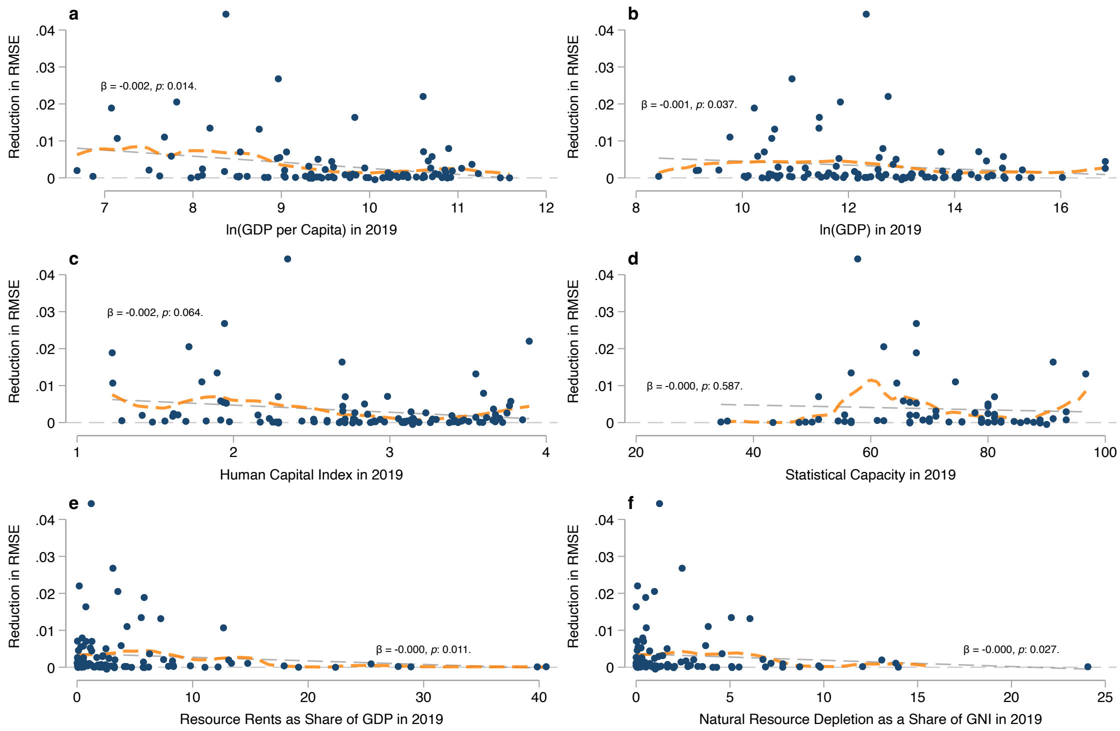
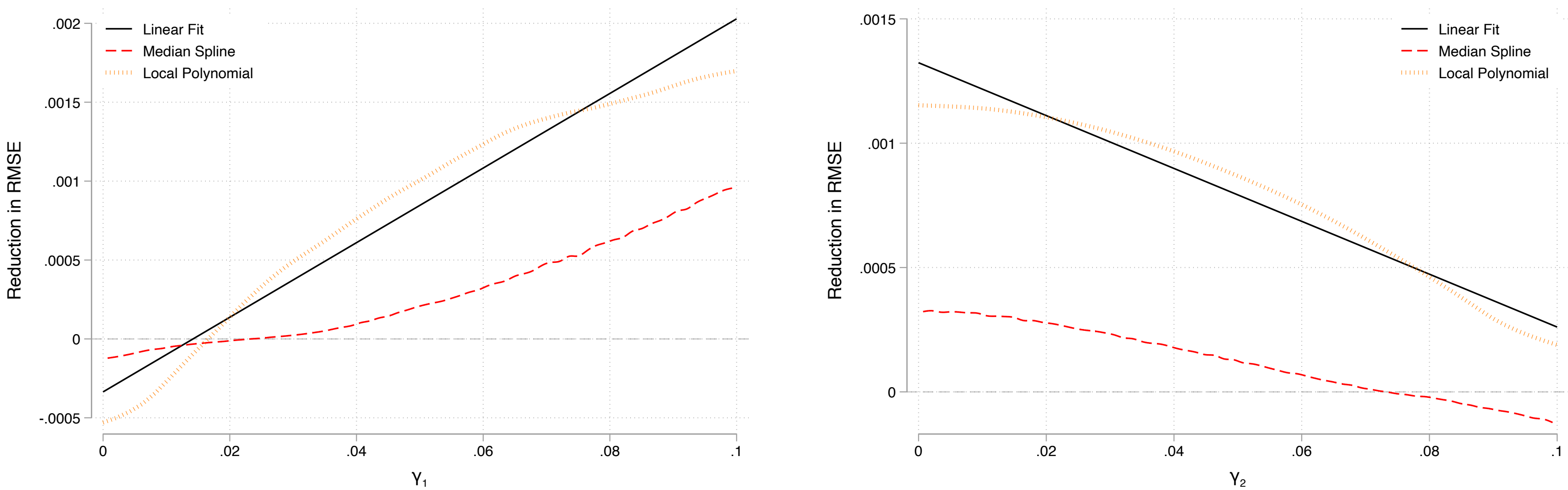


Figure 3: Panels 3a and 3b plot reductions in the RMSE of estimated TFP growth against the (logarithm) of GDP per capita and aggregate GDP at the country level. Panels 3c and 3d plot these improvements against measures of human capital and statistical capacity taken from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). Panels 3e and 3f show RMSE reductions against the share of GDP that comes from resource rents and the share of national income attributable to natural resource depletion, respectively. Dashed lines show fits from simple linear regressions and the overlain text shows the $\hat{\beta}$ coefficient from the linear regression along with the associated p-values. Dashed orange lines show fitted values from a local linear regression (Fan 2018).



(a) Sensitivity of ΔRMSE to Calibrated γ_1

(b) Sensitivity of ΔRMSE to Calibrated γ_2

Figure 4: Linear, median spline, and local polynomial fit lines for ΔRMSE_j at the country level against draws of γ_1 and γ_2 across $N = 10,000,000$ Monte Carlo draws. Fits are performed after winsorizing draws at the 90$^{\text{th}}$ percentile ($|\Delta\text{RMSE}_j| \geq 0.0085$) to reduce the leverage of the right tail of the ΔRMSE_j distribution. The median spline fit is generated by a cubic spline between median values of ΔRMSE_j across observations within each of 70 intervals for the relevant parameter.

References

Acemoglu, Daron, Philippe Aghion, Lint Barrage, and David Hémous. 2023. *Climate change, directed innovation, and energy transition: The long-run consequences of the shale gas revolution.* Technical report. National Bureau of Economic Research. (Cited on page 11).

Aghion, Philippe, and Peter Howitt. 1992. "A Model of Growth through Creative Destruction." *Econometrica* 60 (2): 323–351. (Cited on page 3).

Anderson, Soren T, Ryan Kellogg, and Stephen W Salant. 2018. "Hotelling under pressure." *Journal of Political Economy* 126 (3): 984–1026. (Cited on page 5).

Arellano, Manuel, and Stephen Bond. 1991. "Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations." *The review of economic studies* 58 (2): 277–297. (Cited on page 11).

Arrow, Kenneth, Partha Dasgupta, Lawrence Goulder, Gretchen Daily, Paul Ehrlich, Geoffrey Heal, Simon Levin, Karl-Göran Mäler, Stephen Schneider, David Starrett, et al. 2004. "Are we consuming too much?" *Journal of Economic Perspectives* 18 (3): 147–172. (Cited on page 18).

Barro, Robert J. 1999. "Notes on growth accounting." *Journal of economic growth* 4 (2): 119–137. (Cited on page 3).

Basu, Susanto, Luigi Pascali, Fabio Schiantarelli, and Luis Serven. 2022. "Productivity and the Welfare of Nations." *Journal of the European Economic Association* 20 (4): 1647–1682. (Cited on page 1).

Belloni, Alexandre, Victor Chernozhukov, and Christian Hansen. 2014. "Inference on treatment effects after selection among high-dimensional controls." *Review of Economic Studies* 81 (2): 608–650. (Cited on page 12).

Brandt, Nicola, Paul Schreyer, and Vera Zipperer. 2017. "Productivity measurement with natural capital." *Review of Income and Wealth* 63:S7–S21. (Cited on pages 1, 3, 5, 15, 16, 22).

Bunting, Pete, Ake Rosenqvist, Lara Hilarides, Richard M. Lucas, Tom Thomas, Takeo Tadono, Thomas A. Worthington, Mark Spalding, Nicholas J. Murray, and L-M. Rebelo. 2022. "Global Mangrove Extent Change 1996–2020: Global Mangrove Watch Version 3.0." *Remote Sensing* 14 (15): 3657. https://doi.org/10.3390/rs14153657. (Cited on page 9).

Caselli, Francesco, and James Feyrer. 2007. "The marginal product of capital." *The quarterly journal of economics* 122 (2): 535–568. (Cited on pages 2, 11, 14, 15).

CBO. 2025. *The Budget and Economic Outlook: 2025 to 2035.* Report No. 60870. Accessed 2026-01-19. Congressional Budget Office. https://www.cbo.gov/publication/60870. (Cited on page 1).

CBO. 2026. *The Accuracy of CBO's Budget Projections for Fiscal Year 2025.* 61916. Congressional Budget Office, January. https://www.cbo.gov/system/files/2026-01/61916-budget-accuracy.pdf. (Cited on page 2).

Comin, Diego. 2010. "Total factor productivity." In *Economic growth,* 260–263. Springer. (Cited on page 4).

Conte, Marc N., Ethan T. Addicott, and Maxanne M. Millerhaller. 2025. "Scarcity, willingness to pay for species, and imperfect substitutability with market goods." *Journal of Environmental Economics and Management,* 103192. (Cited on page 24).

Driscoll, John C, and Aart C Kraay. 1998. "Consistent covariance matrix estimation with spatially dependent panel data." *Review of economics and statistics* 80 (4): 549–560. (Cited on page 12).

European Commission and Eurostat. 2025. "Multi-Factor Productivity (MFP) Using National Accounts Data: Methodological Note." https://ec.europa.eu/eurostat/documents/7894008/13933430/Methodological-note-crude-MFP.pdf. (Cited on page 4).

Fan, Jianqing. 2018. *Local polynomial modelling and its applications: monographs on statistics and applied probability 66.* Routledge. (Cited on page 35).

Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer. 2015. "The Next Generation of the Penn World Table." *American Economic Review* 105 (10): 3150–3182. (Cited on pages 4, 7–9, 12, 31, 32, 35).

Fleurbaey, Marc. 2009. "Beyond GDP: The quest for a measure of social welfare." *Journal of Economic literature* 47 (4): 1029–1075. (Cited on page 24).

Freeman, Daan, Robert Inklaar, and W. Erwin Diewert. 2021. "Natural resources and missing inputs in international productivity comparisons." *Review of Income and Wealth* 67 (1): 1–17. (Cited on pages 3, 22).

Hannesson, Rögnvaldur. 2007. "Growth accounting in a fishery." *Journal of environmental economics and management* 53 (3): 364–376. (Cited on pages 1, 5).

Holmlund, Cecilia M, and Monica Hammer. 1999. "Ecosystem services generated by fish populations." *Ecological economics* 29 (2): 253–268. (Cited on page 6).

Hsieh, Chang-Tai, and Peter J. Klenow. 2010. "Development Accounting." *American Economic Journal: Macroeconomics* 2 (1): 207–223. (Cited on page 2).

Jorgenson, Dale W. 2018. "Production and welfare: progress in economic measurement." *Journal of Economic Literature* 56 (3): 867–919. (Cited on page 24).

Kuznets, Simon. 1934. *National Income, 1929–1932.* 73rd Congress, 2nd Session, Senate Document No. 124. Washington, DC: U.S. Government Printing Office. (Cited on page 1).

Lemoine, Derek, Catherine Hausman, and Jeffrey G Shrader. 2026. "Navigating the 'Problem from Hell': A Guide to Climate Damages." *Journal of Economic Literature,* (cited on page 10).

Mankiw, N. Gregory, David Romer, and David N. Weil. 1992. "A contribution to the empirics of economic growth." *The quarterly journal of economics* 107 (2): 407–437. (Cited on pages 2, 15).

Manning, Dale T, and Edward B Barbier. 2025. "Natural capital and aggregate income growth." *Environment and Development Economics* 30 (2): 155–174. (Cited on pages 3, 5, 15, 16, 22).

Manning, Dale T., J. Edward Taylor, and James E. Wilen. 2018. "General equilibrium tragedy of the commons." *Environmental and Resource Economics* 69 (1): 75–101. (Cited on page 18).

Meyer, Peter B., and Michael J. Harper. 2005. "Preliminary Estimates of Multifactor Productivity Growth." *Monthly Labor Review* (June). https://www.bls.gov/opub/mlr/2005/06/art3full.pdf. (Cited on page 4).

Monge-Naranjo, Alexander, Juan M Sánchez, and Raül Santaeulalia-Llopis. 2019. "Natural resources and global misallocation." *American Economic Journal: Macroeconomics* 11 (2): 79–126. (Cited on pages 2, 14–16).

Moscona, Jacob, and Karthik A Sastry. 2023. "Does directed innovation mitigate climate damage? Evidence from US agriculture." *The Quarterly Journal of Economics* 138 (2): 637–701. (Cited on page 11).

Nordhaus, William D, and Edward C Kokkelenberg. 1999. *Nature's Numbers: Expanding the National Economic Accounts to Include the Environment.* (Cited on page 4).

Nordhaus, William D., and James Tobin. 1973. "Is Growth Obsolete?" In *The Measurement of Economic and Social Performance,* edited by Milton Moss, 509–564. National Bureau of Economic Research. (Cited on pages 1, 2, 23, 24).

Organisation for Economic Co-operation and Development. 2001. *Measuring Productivity: OECD Manual.* Paris: OECD Publishing. https://unstats.un.org/unsd/nationalaccount/docs/OECD-Productivity-e.pdf. (Cited on page 4).

Pauly, Daniel, Dirk Zeller, and Maria L. D. Palomares. 2020. *Sea Around Us Concepts, Design and Data.* https://www.seaaroundus.org. Accessed April 5, 2026. (Cited on page 9).

Romer, Paul M. 1990. "Endogenous Technological Change." *Journal of Political Economy* 98 (5, pt. 2): S71–S102. (Cited on page 3).

Roodman, David, Morten Ørregaard Nielsen, James G MacKinnon, and Matthew D Webb. 2019. "Fast and wild: Bootstrap inference in Stata using boottest." *The Stata Journal* 19 (1): 4–60. (Cited on pages 11, 31).

Roosevelt, Theodore. 1908. *Conservation as a National Duty.* Speech delivered at the Governors' Conference on the Conservation of Natural Resources, Washington, DC, May 13–15; reprinted in *Messages and Papers of the Presidents*, vol. 7 (1909), Bureau of National Literature. (Cited on page 1).

Solow, Robert M. 1957. "Technical change and the aggregate production function." *The review of Economics and Statistics* 39 (3): 312–320. (Cited on pages 1, 4).

Törnqvist, Leo, et al. 1936. "The Bank of Finland's consumption price index," (cited on page 18).

U.S. Environmental Protection Agency. 1997. *The Benefits and Costs of the Clean Air Act, 1970 to 1990: Report to Congress.* Technical report EE-0295 (Retrospective Report). Accessed 2026-02-02. U.S. Environmental Protection Agency, Office of Air and Radiation. https://www.epa.gov/sites/default/files/2015-06/documents/contsetc.pdf. (Cited on pages 2, 20).

United Nations. 2009. *System of National Accounts 2008.* Technical report. European Commission, International Monetary Fund, Organisation for the Economic Co-operation and Development, United Nations. (Cited on page 1).

———. 2025. *System of National Accounts 2025.* Technical report. European Commission, International Monetary Fund, Organisation for the Economic Co-operation and Development, United Nations. (Cited on pages 1, 4).

United Nations Statistics Division. 2026. *Beyond GDP: Rethinking Progress in a Changing World — Concept Note.* United Nations Statistics Division, March. https://unstats.un.org/UNSDWebsite/statcom/session_57/side-events/Monday_Seminar_2026_Final_Beyond_GDP_Concept_Note.pdf. (Cited on page 23).

World Bank. 2024a. *The Changing Wealth of Nations 2024: Methodology.* Technical Report. Accessed April 5, 2026. World Bank, October. https://datacatalogfiles.worldbank.org/ddh-published/0042066/10/DR0094582/CWON%202024%20Methodology_10122024.pdf. (Cited on page 8).

———. 2024b. *World Development Indicators.* https://databank.worldbank.org/source/world-development-indicators. Accessed 2026-03-13. World Bank, Washington, DC. (Cited on page 22).

———. 2025. *The Changing Wealth of Nations (CWON) Dataset.* https://data360.worldbank.org/en/dataset/WB_CWON. (Cited on pages 9, 17, 30, 32).